

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – IV

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Part A - Theory:

1. What is ASP.NET Web Forms?

ASP.NET Web Forms is a web application framework developed by Microsoft for building dynamic, interactive, and data-driven websites. It provides a rich collection of **server controls** that simplify web application development by reducing the amount of code required.

Features of ASP.NET Web Forms:

- Event-driven programming model.
- Rich collection of server controls.
- Built-in state management using ViewState.
- Easy integration with databases.
- Support for validation controls.
- Rapid application development.

2. What are Server Controls?

Server Controls are controls that run on the server and generate HTML dynamically. They simplify user interaction and automatically handle client-server communication.

Syntax:

```
<asp:ControlName ID="ControlID" runat="server" />
```

3. Common ASP.NET Server Controls

a) Label Control

The **Label** control is used to display static text or messages on a web page.

Syntax

```
<asp:Label ID="lblName" runat="server" Text="Student Name"></asp:Label>
```

b) TextBox Control

The **TextBox** control allows the user to enter text.

Syntax

```
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
```

c) RadioButton Control

The **RadioButton** control allows the user to select only one option from a group.

Syntax

```
<asp:RadioButton ID="rbMale" runat="server" Text="Male" GroupName="Gender" />
```

d) DropDownList Control

The **DropDownList** control displays a list of options from which the user can select one.

Syntax

```
<asp:DropDownList ID="ddlCourse" runat="server">
  <asp:ListItem>BSc IT</asp:ListItem>
  <asp:ListItem>BSc CS</asp:ListItem>
  <asp:ListItem>BCA</asp:ListItem>
</asp:DropDownList>
```

e) CheckBox Control

The **CheckBox** control allows the user to select or deselect an option.

Syntax

```
<asp:CheckBox ID="chkTerms" runat="server" Text="I Agree" />
```

f) Calendar Control

The **Calendar** control allows users to select a date from a graphical calendar.

Syntax

```
<asp:Calendar ID="Calendar1" runat="server"></asp:Calendar>
```

g) Button Control

The **Button** control is used to perform an action when clicked.

Syntax

```
<asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
```

4. Validation Controls

Validation controls are used to ensure that the user enters valid data before submitting a form.

Types of Validation Controls**a) RequiredFieldValidator**

Ensures that a control is not left empty.

```
<asp:RequiredFieldValidator
ID="rfvName"
runat="server"
ControlToValidate="txtName"
ErrorMessage="Name is required">
</asp:RequiredFieldValidator>
```

b) RangeValidator

Checks whether a value lies within a specified range.

c) CompareValidator

Compares the value of one control with another control or a fixed value.

d) RegularExpressionValidator

Validates the input using a regular expression (e.g., Email, Mobile Number).

e) CustomValidator

Allows custom validation logic written by the programmer.

f) ValidationSummary

Displays all validation error messages in one place.

5. AdRotator Control

The **AdRotator Control** displays different advertisements or images randomly from an XML file. It is commonly used to display advertisements, announcements, offers, or promotional banners.

Advantages

- Displays advertisements automatically.
- Randomly rotates images.
- Reads advertisement details from an XML file.
- Easy to maintain and update.

Syntax

```
<asp:AdRotator  
ID="AdRotator1"  
runat="server"  
AdvertisementFile="Ads.xml">  
</asp:AdRotator>
```

6. Advantages of ASP.NET Server Controls

- Easy to use and configure.
- Supports event-driven programming.
- Reduces coding effort.
- Provides automatic state management.
- Supports built-in validation.
- Improves application security.
- Generates browser-compatible HTML automatically.

7. Applications

- Student Registration System
- Online Admission Portal
- Employee Registration System
- Online Examination System
- Library Management System
- E-Commerce Registration Forms
- Hospital Appointment System

Part B - Program

- a) Create an ASP.NET Web Forms application using simple server controls.
- a. Design a Student Registration Form using the following ASP.NET controls:
- i. Label and TextBox to accept Student Name
 - ii. RadioButton to select Gender (Male/Female)
 - iii. DropDownList to select Course (BSc IT, BSc CS, BCA)
 - iv. CheckBox for accepting Terms & Conditions
 - v. Calendar control to select Date
 - vi. Button control to submit the form
 - vii. Label control to display the output

Algorithm:

- Step 1:** Start the program
- Step 2:** Display the form with input controls (TextBox, RadioButtonList, DropDownList, CheckBox, Calendar, Button)
- Step 3:** Accept student name from the TextBox
- Step 4:** Accept gender from the RadioButtonList
- Step 5:** Accept course from the DropDownList
- Step 6:** Accept date from the Calendar
- Step 7:** On Submit button click, retrieve all values and store in variables
- Step 8:** Display the entered details in Label
- Step 9:** Stop further input processing
- Step 10:** End the program

Code:

Default1.aspx :

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Default3.aspx.cs" Inherits="Default3" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
  <title></title>
</head>
<body>
  <form id="form1" runat="server">
    <div>
      <asp:Label ID="Label1" runat="server" Text="Student Name :"></asp:Label>
      &nbsp;
      <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>

      <br />

      <asp:RadioButtonList ID="RadioButtonList1" runat="server">
        <asp:ListItem Value="M">Male</asp:ListItem>
        <asp:ListItem Value="F">Female</asp:ListItem>
        <asp:ListItem Value="O">Other</asp:ListItem>
      </asp:RadioButtonList>

      <br />

      <asp:DropDownList ID="DropDownList1" runat="server">
        <asp:ListItem>BSc IT</asp:ListItem>
        <asp:ListItem>BSc CS</asp:ListItem>
        <asp:ListItem>BCA</asp:ListItem>
      </asp:DropDownList>

      <br />

      <asp:CheckBox ID="CheckBox1" runat="server" Text="I Accept Terms and Conditions." />

      <br />
    </div>
  </form>
</body>
</html>
```

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

2 references
public partial class Default3 : System.Web.UI.Page
{
    0 references
    protected void Page_Load(object sender, EventArgs e)
    {
    }

    0 references
    protected void Button1_Click(object sender, EventArgs e)
    {
        string name = TextBox1.Text;
        string gender = RadioButtonList1.Text;
        string course = DropDownList1.Text;
        string date = Calendar1.SelectedDate.ToString();

        Label2.Text = name + "<br/>" + gender + "<br/>" + course + "<br/>" + date;
    }
}

```

Student Name :

☐ Male
☐ Female
☐ Other

BSc IT

☐ I Accept Terms and Conditions.

< July 2026 >

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Submit

Output:

Student Name :

☒ Male
☐ Female
☐ Other

BSc IT

☒ I Accept Terms and Conditions.

≤ July 2026 ≥

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Submit

Siddhesh Kamble
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b) Create a Registration form to demonstrate use of various Validation controls.

Algorithm:

- Step 1:** Start the program
- Step 2:** Display the form with Labels, TextBoxes, and Button
- Step 3:** Accept Name input and validate using RequiredFieldValidator
- Step 4:** Accept Age input and validate using RangeValidator (18–99)
- Step 5:** Accept Password and Confirm Password inputs
- Step 6:** Validate passwords using CompareValidator
- Step 7:** On Submit button click, check all validations
- Step 8:** If valid, accept input and end the program

Code:

Default2.aspx :

```

<!-- Page Language="C#" AutoEventWireup="true" CodeFile="Default2.aspx.cs" Inherits="Default2" -->
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<div>
<asp:Label ID="Label1" runat="server" Text="Name "></asp:Label>
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
<asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server" ControlToValidate="TextBox1" ErrorMessage="Name can not be empty ."></asp:RequiredFieldValidator>
<br />
<asp:Label ID="Label2" runat="server" Text="Age "></asp:Label>
<asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
<asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox2" ErrorMessage="Age should be in between 18 and 99" MaximumValue="99" MinimumValue="18"></asp:RangeValidator>
<br />
<asp:Label ID="Label3" runat="server" Text="Password "></asp:Label>
<asp:TextBox ID="TextBox3" runat="server" TextMode="Password"></asp:TextBox>
<br />
<asp:Label ID="Label4" runat="server" Text="Confirm Password "></asp:Label>
<asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
<asp:CompareValidator ID="CompareValidator1" runat="server" ControlToCompare="TextBox3" ControlToValidate="TextBox4" ErrorMessage="Password not match ."></asp:CompareValidator>
<br />
<asp:Button ID="Button1" runat="server" Text="Submit " />
<asp:AdRotator ID="AdRotator1" runat="server" AdvertisementFile="--/XMLFile.xml" />
</div>
</form>
</body>
</html>

```

Output:

c) Create Web Form to demonstrate use of Adrotator Control.

Algorithm:

- Step 1:** Start the program
- Step 2:** Display the web form with input controls and validation controls
- Step 3:** Accept Name and validate using RequiredFieldValidator
- Step 4:** Accept Age and validate using RangeValidator (18–99)
- Step 5:** Accept Password and Confirm Password and validate using CompareValidator
- Step 6:** Load XML file containing advertisement details
- Step 7:** Display images with links, alternate text, height, and width from XML data
- Step 8:** On Submit button click, validate all inputs and end the program

Code:

Default3.aspx :

```

<?xml version="1.0" encoding="utf-8" ?>
<DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<div>

<asp:Label ID="Label1" runat="server" Text="Name "></asp:Label>
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
<asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server" ControlToValidate="TextBox1" ErrorMessage="Name can not be empty ."></asp:RequiredFieldValidator>
<br />
<asp:Label ID="Label2" runat="server" Text="Age "></asp:Label>
<asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
<asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox2" ErrorMessage="Age should be in between 18 and 99" MaximumValue="99" MinimumValue="18"></asp:RangeValidator>
<br />
<asp:Label ID="Label3" runat="server" Text="Password "></asp:Label>
<asp:TextBox ID="TextBox3" runat="server" TextMode="Password"></asp:TextBox>
<br />
<asp:Label ID="Label4" runat="server" Text="Confirm Password "></asp:Label>
<asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
<asp:CompareValidator ID="CompareValidator1" runat="server" ControlToCompare="TextBox3" ControlToValidate="TextBox4" ErrorMessage="Password not match ."></asp:CompareValidator>
<br />
<asp:Button ID="Button1" runat="server" Text="Submit " />
<br />
<asp:AdRotator ID="AdRotator1" runat="server" AdvertisementFile="~/XMLFile.xml" />

</div>
</form>
</body>
</html>

```

XMLFile.xml :

```

<?xml version="1.0" encoding="utf-8" ?>
<Advertisements>
<Ad>
<ImageUrl>images/img1.png</ImageUrl>
<NavigatorUrl>https://www.google.com</NavigatorUrl>
<AlternateText>Tiger Image</AlternateText>
<Height>50</Height>
<Width>50</Width>
</Ad>
<Ad>
<ImageUrl>images/img2.png</ImageUrl>
<NavigatorUrl>https://www.google.com</NavigatorUrl>
<AlternateText>Cat Image</AlternateText>
<Height>100</Height>
<Width>100</Width>
</Ad>
</Advertisements>

```

The screenshot shows the rendered web form. It contains four text input fields: 'Name', 'Age', 'Password', and 'Confirm Password'. Each field has a validation error message displayed next to it: 'Name can not be empty .', 'Age should be in between 18 and 99', 'Password not match .', and 'Password not match .'. Below the input fields is a 'Submit' button. At the bottom of the form, there is an AdRotator control displaying an advertisement for a tiger image with a height of 50 and width of 50.

Output:

Name
Age
Password
Confirm Password



Name
Age
Password
Confirm Password

